Linguistic Universals as Cognitive Universals: Causal Coding, Quantale Weakness and Categorial Correspondences

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Abstract

This paper formulates a category-theoretic and quantale-weakness correspondence between two threads of work: a TyLAA/TUG account of real-world linguistic universals and a causal-coding account of modular continual learning. The two central claims are:

$$\boxed{\begin{array}{c}\text{linguistic universals are direct images of cognitive invariants}\\ \text{along a weakest, quantale-enriched externalization correspondence}\end{array}}$$

and conversely

$$\boxed{\begin{array}{c}\text{cognitive universals are weakest pullback explanations}\\ \text{of stable linguistic subobjects.}\end{array}}$$

In ordinary language: robust cross-linguistic patterns are the observable shadows of stable patterns in the cognitive substrate, transported into language by the machinery that externalizes cognition; and conversely, the best cognitive explanation of any given typological pattern is the weakest cognitive structure that suffices to make the pattern stable. "Weakest" is given an algebraic meaning, not just a metaphorical one.

We make this precise by introducing three quantale-enriched categories: a category Cog_Q of causally factorized cognitive learners, a category Frame_Q of semantic frames, and a category Lang_Q of TUG trace quotients under typological observation interfaces. The cognition-to-language map is a lax symmetric monoidal Q-enriched functor

$$\Lambda = \mathcal{E} \circ \mathcal{K} : \mathsf{Cog}_Q \longrightarrow \mathsf{Lang}_Q,$$

in which $\mathcal{K}$ extracts semantic-frame structure from a cognitive substrate and $\mathcal{E}$ externalizes frames into linguistic cores. When a single cognitive system can support multiple externalizations, or one linguistic core can be supported by several cognitive substrates, the more general object is a Q-enriched profunctor $\mathcal{P} : \mathsf{Cog}_Q^{op} \times \mathsf{Lang}_Q \to Q$. Within this setting, linguistic universals are subobjects (or predicates) in Lang_Q; cognitive explanations are pullbacks of those subobjects along Λ; and a quantale-valued weakness criterion selects among them by penalizing unnecessary cost, confusion, and double-counting.

In this language, the five informal correspondences linking the TUG paper to the causal-coding paper become theorems or theorem-shaped claims. The strongest is a *monoidal transport theorem*: cognitive updates supported on disjoint causal factors have linguistic images that are swap-equivalent in the TUG trace quotient, and cognitive updates that nearly commute have linguistic images whose swap defect is bounded by the image of the cognitive commutator defect. This single enriched monoidal fact unifies the TUG trace-factorization theorem with the causal-continual-learning commutator theorem. The remaining four correspondences become a closure transport theorem, a sparse-energy pushforward result, a stability-rate transport theorem, and an exact-and-approximate gluing theorem for context-indexed laws. Quantale weakness supplies the selection principle that ties everything together: causal explanations dominate redundant correlation-only explanations because they introduce the weakest sufficient mediator and avoid double-counting the same dependency through multiple paths.

We additionally extend the framework from morphosyntactic universals to *semantic-evolution universals*, by inserting a dynamic lexical-semantic layer between frames and observed linguistic form. Empirical regularities reported by Guo et al. [**?**] for word-meaning evolution across many languages frequency assortativity, characteristic clustering velocity profiles, persistent local temporal dynamics, and Taylor's law for new-entity counts are then recovered as observable shadows of attention-weighted, hierarchical, context-local, and bursty mechanisms of concept formation in the cognitive substrate. This adds a sixth class to the taxonomy of universals (semantic growth-scaling laws) and shows that the same enriched correspondence governs how cognitive-cultural meaning space *grows* as well as how linguistic form is constrained.

Contents

1 Introduction

1.1 The main thesis

The TUG-based paper on linguistic universals [**?**] extracts five mathematical conclusions from the corrected typological record. The causal-coding papers on continual learning [**?**, **?**] derive, from very different first principles, five architectural features that any continual learner must have if it is to learn many contexts over a lifetime without catastrophic forgetting. The two lists line up closely, item for item. This paper argues that the alignment is not coincidental but structural. The argument takes the form of a structured map between three categories,

$$\text{cognitive causal systems} \quad \longrightarrow \quad \text{semantic frames} \quad \longrightarrow \quad \text{linguistic TUG cores,}$$

together with a weakness order that selects among possible cognitive explanations of any given linguistic pattern. In this setup, the central claim becomes a single statement, with five theorems or theorem-shaped propositions falling out of it as special cases:

$$\boxed{\begin{array}{c}\text{linguistic universals are direct images of cognitive invariants}\\ \text{along a weakest, quantale-enriched externalization correspondence}\end{array}}$$

and, in the inverse direction,

$$\boxed{\begin{array}{c}\text{cognitive universals are weakest pullback explanations}\\ \text{of stable linguistic subobjects.}\end{array}}$$

These sentences are deliberately strong, but they should not remain Delphic. The next subsection unpacks them.

1.2 Unpacking the two boxed claims

A *cognitive invariant* is a pattern in the cognitive substrate that remains stable across contexts. At the level of abstraction relevant here, examples include person-role frames, number frames, argument-role frames, causal-event frames, and, more globally, the modular decomposition of the substrate that prevents one context of learning from destroying another. A *linguistic universal* is an observable regularity in the TUG core of human languages: a person hierarchy, a number hierarchy, a case hierarchy, a word-order coherence tendency, or a context-local grammatical law.

A *direct image* is the mathematical version of pushing a pattern forward through a structure-preserving map. If a cognitive pattern is stable inside the mind, and if the language-production system preserves the relevant structure (at least up to controlled slack), then the image of that cognitive pattern becomes a linguistic regularity. A *pullback* is the inverse construction: it takes a linguistic pattern and asks what structure must have existed upstream in order for the pattern to be observed downstream. A *weakest pullback explanation* is the least overcommitted upstream explanation: the one that explains the linguistic pattern while positing the fewest unnecessary distinctions, couplings, hidden variables, or double-counted correlations.

The word *quantale* supplies the quantitative Occam principle. A quantale is an algebraic structure that lets us combine and compare values such as weakness, cost, confidence, confusion, and double-counting in a single ordered framework. It allows us to say algebraically, not just informally, that one explanation is weaker than another because it preserves more indistinction while using less cost and fewer redundant dependencies. This is the version of weakness developed in [**?**]; it generalizes the TyLAA/TUG idea of choosing the weakest red operational system consistent with a blue type interface [**?**, **?**], and composes that idea with causal-coding weakness and with the independence-versus-dependency Pareto frontier of [**?**].

So the first boxed claim says: a linguistic universal is what shows up downstream when a stable cognitive pattern is pushed through the cognition-to-language pipeline; and "weakest" picks out the version of that pipeline that introduces the least extra structure. The second claim says: from the linguistic side, a stable typological pattern is evidence for some upstream cognitive structure, but only for the weakest such structure that the data force us to posit.

1.3 Why category theory is the right language here

The core problem is not merely to compare two sets of features. Linguistic universals live in a space of derivations, observation interfaces, quotients, symmetries, context restrictions, and diachronic transitions. Cognitive universals live in a space of causal modules, update endomorphisms, semantic frames, context supports, and stability gradients. The map between them is not a function from one list to another; it has to preserve composition, modularity, locality, and weakness. Category theory is designed for exactly this kind of situation: it is the natural language for talking about how structured worlds map into other structured worlds.

For readers without a category-theoretic background, the relevant working vocabulary is short.

- A **category** is a world of objects together with structure-preserving arrows between them. Composition of arrows is associative and each object has an identity arrow.
- A **functor** is a structure-preserving map between such worlds: it sends objects to objects, arrows to arrows, and preserves composition and identities.
- A **monoidal category** is a category in which objects can also be composed side by side via a tensor product. Side-by-side composition represents modular factorization: an object of the form $A \otimes B$ is a system that decomposes into two pieces.
- A **lax monoidal functor** preserves side-by-side composition only up to a controlled slack. This matters because cognition-to-language externalization preserves modularity only approx-

imately: one cognitive module may externalize through several morphological channels, or several cognitive modules may share a single phonological channel.

- An **enriched category** (or, in our lighter form, a *weighted* category) replaces yes/no arrows by arrows valued in an external structure such as a quantale. This lets each morphism carry an explicit grade: how much weakness it preserves, how much cost it incurs, how much confusion it introduces.
- A **pullback** is the categorical preimage of a pattern. It tells us what upstream structure is required for a downstream pattern to be possible.
- A **direct image** pushes an upstream pattern forward into the downstream world along a functor; it is left adjoint to pullback when both make sense.
- An **adjunction** formalizes a pair of translations that fit together as two halves of one process. Here, externalization (frame-to-language) and interpretation (language-to-frame) are modeled as adjoint operations between semantic frames and linguistic forms; this is the categorical version of the production/comprehension duality.

These notions are not decorative. They turn the central thesis from a suggestive comparison into a precise transport theorem.

1.4 The three-layer correspondence

The basic diagram of the paper is:

$$\mathsf{Cog}_Q \xrightarrow{\ \mathcal{K}\ } \mathsf{Frame}_Q \xrightarrow{\ \mathcal{E}\ } \mathsf{Lang}_Q.$$

Here Cog_Q is the category of causal-coding cognitive systems, Frame_Q is the category of semantic frames, and Lang_Q is the category of TUG linguistic cores. The functor $\mathcal{K}$ extracts from a cognitive system the frames its causal modules can support, and $\mathcal{E}$ externalizes those frames into linguistic structure. The composite

$$\Lambda := \mathcal{E} \circ \mathcal{K} : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$$

is the cognition-to-language correspondence; this is the functor through which direct images and pullbacks are taken.

The intermediate frame layer is essential. It would be too rigid to map directly from neural or cognitive modules to surface linguistic features: between them lie semantic frames, including event frames, force-dynamic frames, argument-role frames, person/number frames, possession frames, attention frames, and other frame-like cognitive schemas. The general theory of general intelligence in [**?**] treats mind as a network of patterns and emphasizes cognitive synergy among different kinds of cognitive process; the frame category here is a formal bridge between that patternist/cognitive-synergy picture and the TUG machinery of grammar space. In this sense the diagram above is not just a piece of mathematics but a concrete claim about how cognition and language are organized: a layer of causal cognitive structure, a layer of frame-like meaning patterns, and a layer of TUG-quotiented linguistic form.

Figure **??** is an illustrative summary. Concrete examples of objects in each layer are shown inside the boxes. The solid forward arrows are the functors $\mathcal{K}$ and $\mathcal{E}$; the dashed forward curve indicates the composite direct-image action of Λ on cognitive invariants, sending them to linguistic universals; the dashed reverse curve indicates the inverse direction in which a stable linguistic universal pulls back along Λ to its weakest cognitive explanation.

A time-indexed dynamic version of this diagram, with an additional lexical-semantic growth layer between frames and observed embeddings, is developed in Section **??** and applied to the four embedding-space regularities of [?]. The static and dynamic correspondences are governed by the same enriched-categorical machinery; the dynamic version simply tracks how cognitive-cultural meaning space grows over historical time as well as how stable cognitive structure constrains linguistic form.

1.5 The empirical motivation

The TUG paper [?] joins a TyLAA-based formal account of grammar cores with large-scale statistical evidence about proposed implicational universals [?]. After genealogical and areal correction, the surviving empirical pattern is striking and structured: hierarchical universals survive correction best; narrow word-order universals survive moderately; broad cross-module universals survive poorly unless they involve explicit mediators; universalhood is graded rather than binary; and many genuine universals turn out to be context-indexed local laws rather than sweeping global ones.

The causal-coding papers, starting from continual-learning considerations independent of any linguistic data, derive the same kind of architecture: a protected stable substrate, modular factorization, sparse cross-module coupling, graded kernel-shell stability, and context-conditioned routing [?, ?]. The empirical claim of the present paper is that this match is not a coincidence. If language is in fact carried by a cognitive architecture that has to learn many contexts over a lifetime without catastrophic forgetting, then the typological footprint left by that architecture should look exactly like the corrected linguistic-universals record. The categorical formulation then turns this empirical claim into a structural theorem about what the typological record can be evidence *for*.

A second, complementary, empirical anchor comes from the statistical structure of word-meaning evolution. Recent work by Guo et al. [?] reports four regularities in empirical word-embedding spaces, drawn from English and 21 additional languages: frequency assortativity, characteristic clustering velocity profiles, persistent local temporal dynamics in the appearance of new entities, and Taylor's law for the fluctuations of new-entity counts. The four regularities are reproduced by a class of directed preferential placement generative models and not by simpler uniform or purely preferential alternatives. These regularities are not merely facts about surface text: they describe how a semantic or epistemic space *grows* over historical time. Section **??** below shows that they fit naturally into the same enriched cognition-language correspondence, as the lexical-semantic shadow of cognitive-cultural growth dynamics with attention-weighted, hierarchical, context-local, and bursty structure.

1.6 Contributions

The contributions of this paper are as follows.

1. It defines Q-enriched categories Cog_Q, Frame_Q, and Lang_Q, together with a lax monoidal correspondence $\Lambda = \mathcal{E} \circ \mathcal{K}$ factoring through frames.

2. It treats linguistic universals as subobjects in Lang_Q, cognitive invariants as subobjects in Cog_Q, and the cognition-language relation as direct image and pullback between subobject lattices.

3. It defines a category $\mathrm{Exp}(U)$ of explanations of a linguistic universal U, and selects the weakest explanation using quantale weakness in a product quantale that grades indistinction, fit, cost, confusion, and double-counting.

three_layers.png

Figure 1: The three-layer correspondence between cognition, semantic frames, and language. Concrete examples in each layer are shown inside the corresponding boxes. Solid arrows are the functors $\mathcal{K}$ (extract) and $\mathcal{E}$ (externalize); the dashed forward curve at the bottom is the composite direct-image action of $\Lambda = \mathcal{E} \circ \mathcal{K}$, sending cognitive invariants to linguistic universals; the dashed reverse curve at the top is the inverse pullback direction, sending a stable typological pattern back to its weakest cognitive explanation.

4. It proves a monoidal transport theorem showing that causal modularity maps to TUG modularity and that commutator defects on the cognitive side bound swap defects on the linguistic side.

5. It formulates linguistic mediators as residuals: weakest contexts in a behavior quantale that make a cross-module implication true.

6. It presents the five informal correspondences between TUG conclusions and causal-coding architectural features as closure transport, factorization, sparse-energy pushforward, stability-rate transport, and context-indexed gluing theorems, each stated under explicit assumptions.

7. It extends the framework to semantic-evolution universals by inserting a dynamic lexical-semantic layer $\mathsf{LexSemDyn}_Q$ between frames and observed linguistic form, recovering the four embedding-space regularities of Guo et al. [?] as observable shadows of attention-weighted, hierarchical, context-local, and bursty cognitive growth, and adding a sixth class (semantic growth-scaling) to the taxonomy of universals.

2 Background: TUG, causal coding, and quantale weakness

2.1 TUG as the weakest typological grammar core

The TUG formalism begins with an operational grammar G_ℓ for a language ℓ. The derivations of that grammar form a space $\mathrm{Deriv}(G_\ell)$. One then specifies three pieces of additional structure: an observation interface $\mathcal{O}$ describing what is to be considered observable, an independence relation $\perp$ describing which actions can be commuted past each other without observable effect, and a symmetry group Γ describing internal renamings. For typology, the observation interface must record at least logical form, phonological/externalization information, and morphosyntactic information:

$$\mathcal{O}_{\mathrm{typ}} = (\Sigma_{\mathrm{LF}}, \Sigma_{\mathrm{PF}}, \Sigma_{\mathrm{Morph}}).$$

The typological TUG core is the quotient

$$C_\ell := \mathrm{TUG}_{\mathcal{O}_{\mathrm{typ}}}(G_\ell) := \mathrm{Deriv}(G_\ell)/{\equiv_{\perp,\Gamma,\mathcal{O}_{\mathrm{typ}}}},$$

where $\equiv_{\perp,\Gamma,\mathcal{O}_{\mathrm{typ}}}$ is the maximal sound congruence generated by swapping adjacent independent actions, applying the symmetries Γ, and identifying derivations that the observation interface cannot distinguish.

For non-specialists: the quotient implements the principle that, if two derivations are observationally indistinguishable through the typological interface, then the grammar core should not keep them apart. The core C_ℓ is therefore the weakest operational system that still preserves everything typology can observe. This is the natural-language analogue of the TyLAA principle that, given a blue type/proof interface, one chooses the weakest red operational explanation consistent with it [?, ?].

2.2 The five TUG universals conclusions

The TUG-based linguistic-universals paper [?] extracts five mathematical conclusions from the empirical pattern of which proposed universals survive correction.

C1. Hierarchical universals are closure systems. A hierarchy of implications generates a closure operator cl_R on a feature set. Admissible paradigms are fixed points of that operator, equivalently order ideals after quotienting mutual implication.

C2. Cross-module universals require mediation. If two grammatical modules are independent in the TUG sense, the trace quotient factors; no nontrivial implication can cross from one factor to another unless a mediator breaks the factorization.

C3. Narrow word-order universals are low-frustration externalization laws. Word-order regularities are modeled not by one global head-direction bit but by a sparse signed graph of local externalization choices.

C4. Universalhood is graded by dynamical stability. A proposed universal should be assigned a stability coefficient derived from transition rates rather than a binary supported/unsupported label.

C5. Many genuine universals are context-indexed local laws. Broad unconditional claims often fail because they pool over distinct grammatical contexts in which real local laws hold.

These conclusions already have a categorical flavor. Closure systems are algebraic subobject structures; factorization is a monoidal statement; low-frustration word order is an energy pushforward problem; stability is a functorial comparison of dynamics; context-indexed laws are sheaf-like gluing problems. The rest of the paper makes this explicit.

2.3 Causal coding and the General Causal-Continual-Learning theorem

Causal coding starts from a continual-learning problem. A learner encounters a stream of contexts $c \in C$, and each context c induces an update operator U_c on the learner's internal state. If every context rewrites the same entangled bundle of parameters, catastrophic forgetting is unavoidable: each new context overwrites what previous contexts had encoded. If, instead, the internal state factors into causal modules and each context updates mostly the modules it actually needs, then the corresponding updates approximately commute, and sequential learning across many contexts approximates joint learning across all of them at once.

In its categorical form, the General Causal-Continual-Learning (CCL) theorem considers a monoidal object representing the learner's state,

$$A \cong A_1 \otimes \cdots \otimes A_n,$$

together with update endomorphisms

$$U_c : A \to A$$

indexed by contexts c, each with a support $S_c \subseteq \{1, \ldots, n\}$ listing the modules it is allowed to touch. Outside its support, an update is the identity. The theorem then states that updates whose supports are disjoint commute up to the canonical monoidal symmetry:

$$S_c \cap S_d = \emptyset \quad \Longrightarrow \quad U_c \circ U_d \cong U_d \circ U_c.$$

The approximate version of the theorem measures deviations from disjointness using a metric or quantale-valued defect, and bounds total path-dependence by a sum over genuinely confusing context pairs [**?**]. This commutator/factorization theorem is the cognitive-learning counterpart of the TUG trace-factorization theorem: in both cases, structural independence forces algebraic independence, and approximate independence forces approximate algebraic independence.

2.4 HBCML and kernel-shell stability

HBCML and ColBaC make the CCL theorem constructive [?]. They propose a two-level architecture:

- a sparse top-level controller selecting a subset of modules or columns for each context;
- internal probabilistic structure within each column, distinguishing protected hard-kernel material, reusable inner-shell structure, semi-general middle-shell structure, and task-local outer-shell residue.

The hard kernel is non-prunable; inner shells are difficult to perturb; middle shells are adaptable; outer shells are disposable. This is a natural architectural correlate of the graded stability coefficient proposed for universals in [?].

2.5 Quantale weakness

The weakness framework [?] generalizes Occam's razor from syntactic shortness ("shorter description = better") to semantic undercommitment ("fewer unnecessary distinctions = better"). In its simplest form, a hypothesis is weak if it identifies many pairs of objects that did not need to be distinguished. In the quantale formulation, one chooses a commutative quantale $(Q, \leq, \otimes, e, \bigvee)$, a set X of objects, and a valuation $\mu : X \to Q$ recording how important each object is. For a relation $H \subseteq X \times X$ interpreted as indistinction, one then defines

$$w(H) := \bigvee_{(x,y)\in H} \mu(x) \otimes \mu(y).$$

Larger $w(H)$ means the relation treats more important pairs as indistinct, which means it makes fewer unnecessary distinctions, and therefore generalizes more. By varying the quantale and valuation one recovers Bennett-style set weakness, probabilistic weakness, logical-entropy / graphtropy-style weakness, and several other variants as instances of the same construction.

For the present paper we need a product quantale that tracks several desiderata at once.

Definition 2.1 (Explanatory weakness quantale)**.** Let

$$Q = Q_{\text{ind}} \times Q_{\text{fit}} \times Q^{op}_{\text{cost}} \times Q^{op}_{\text{conf}} \times Q^{op}_{\text{dc}},$$

where Q_{ind} measures indistinction or generalization, Q_{fit} measures empirical fit or confidence, Q_{cost} measures computational or description cost, Q_{conf} measures commutator/confusion cost, and Q_{dc} measures double-counting. The superscript *op* reverses the order in the cost-like components, so that lower cost, lower confusion, and lower double-counting all count as "better". The product is ordered componentwise, with componentwise tensor products and joins inherited from the factors.

Remark 2.2. The notation $Q^{op}_{\bullet}$ is used informally here. Reversing the order of a quantale does not always give a quantale in the strict sense, because joins may fail to be preserved. The use of $Q^{op}_{\bullet}$ above should be understood as follows: each component is an ordered semiring used to compare candidates componentwise, with the cost-like components compared with reversed orientation. All theorems below depend only on the resulting product preorder and on the existence of joins/meets within each component on the relevant sublattices, not on the strict quantale structure of the overall product.

The interpretation is straightforward: an explanation is better, hence weaker in the generalized Occam sense, if it preserves more admissible indistinction, fits the data at least as well, uses less cost, induces less context confusion, and double-counts fewer dependencies.

2.6 The independence/dependency Pareto frontier

The tradeoff paper [?] argues that practical intelligence lives on a Pareto frontier between two routes:

Route A: transform the representation so that channels become more independent;

Route B: explicitly model dependencies among channels.

Causal explanations are favored in this framework because they often identify a small number of genuine mediators instead of simulating the same dependence through many overlapping correlations. In the quantale above, the latter incurs a larger DC component and often a larger Conf component. Thus the weakness principle does not merely say "prefer simpler explanations" in prose; it supplies an algebraic order in which causal explanations can dominate redundant correlation-only explanations.

3 The three enriched categories

We now define the categorical setting. The definitions are intentionally somewhat abstract. They are meant to isolate the structure needed for the theorems rather than to encode every implementation detail of every possible cognitive or linguistic system.

3.1 A note on enriched and weighted categories

For technical convenience we use a lightweight version of enrichment. A Q-weighted category is an ordinary category whose arrows additionally carry grades in the quantale Q, with composition respecting those grades in the natural way. When Q is viewed as a one-object monoidal category, this is a familiar form of enrichment in the sense of Kelly [?]; when the arrows are ordinary maps annotated with explicit defect bounds, it is often more readable to call the result a weighted category and treat the grades as tracking quantitative properties rather than full hom-objects.

Definition 3.1 (Q-weighted category)**.** A Q-weighted category $\mathcal{C}$ consists of an ordinary category, together with a grade function

$$\omega_{X,Y} : \mathcal{C}(X,Y) \to Q$$

such that

$$e \leq \omega(\mathrm{id}_X), \qquad \omega(g) \otimes \omega(f) \leq \omega(g \circ f)$$

whenever $f : X \to Y$ and $g : Y \to Z$. We read $\omega(f)$ as the weakness/quality grade of the morphism f; the product quantale order is chosen so that *larger* grades mean "better".

For readers unfamiliar with this language: in a strict category an arrow either exists or does not, full stop. Here, an arrow can exist with a quantitative quality profile, recording how much structure it preserves, how much error it introduces, how much it costs, and so on. The composition inequality says that composing two arrows can only make the combined quality worse than the tensor of the separate qualities; you do not gain quality by chaining steps.

3.2 The cognitive category Cog_Q

The category Cog_Q is meant to capture the kind of cognitive system to which the causal-coding theorem applies: a learner whose internal state factors into causal modules, whose contexts induce update operators with bounded support, and whose updates can be graded by weakness, cost, and confusion.

Definition 3.2 (Causal-coding cognitive object)**.** An object $A \in \mathsf{Cog}_Q$ is a tuple

$$A = (\Theta_A \cong \Theta_{A,1} \otimes \cdots \otimes \Theta_{A,n_A}, C_A, \{U_c : A \to A\}_{c \in C_A}, \{S_c \subseteq \{1, \ldots, n_A\}\}_{c \in C_A}, \mu_A),$$

where:

- Θ_A is the internal state or parameter object;
- $\Theta_{A,i}$ are the causal modules into which Θ_A factors;
- C_A is a set or category of contexts;
- U_c is the update endomorphism induced by context c;
- S_c is the support of context c, i.e. the set of modules that U_c is allowed to touch nontrivially;
- μ_A is a valuation assigning weakness, cost, and importance grades to states, modules, frames, or update traces.

A morphism in Cog_Q is a causal simulation: it maps one causal learner into another while preserving the context-conditioned update structure up to controlled defect.

Definition 3.3 (Causal simulation morphism)**.** A morphism

$$f : A \to B$$

in Cog_Q consists of a map of state objects $f_\Theta : \Theta_A \to \Theta_B$ and a map of contexts $f_C : C_A \to C_B$, such that for each context $c \in C_A$ the square

$$\begin{tikzcd} \Theta_A \arrow[r, "U_c"] \arrow[d, "f_\Theta"'] & \Theta_A \arrow[d, "f_\Theta"] \\ \Theta_B \arrow[r, "U_{f_C(c)}"'] & \Theta_B \end{tikzcd}$$

commutes up to a quantale-valued defect

$$\delta_c(f) := d_B\big(f_\Theta \circ U_c, U_{f_C(c)} \circ f_\Theta\big).$$

The grade $\omega(f) \in Q$ is then obtained by aggregating the indistinction preserved by f, the empirical fit, the cost, the confusion defects $\delta_c(f)$, and the double-counting cost.

The intuitive content is just this: a morphism $A \to B$ says that the second cognitive system can simulate the first while preserving its causal update structure. The square says that it should not matter whether one first updates in context c and then maps forward into B, or first maps forward into B and then updates in the corresponding context $f_C(c)$. The defect $\delta_c(f)$ measures the failure of this equality, and the grade $\omega(f)$ packages all the various qualities of the simulation into a single quantale element.

3.3 The semantic-frame category Frame_Q

The frame layer is what makes the cognition-language map well-typed. Cognitive modules are usually too low-level, and linguistic features too surface-level, for any direct functor to do useful work. Frame structures, in the sense of frame semantics and related cognitive-science traditions, sit naturally between them: they are abstract enough to be shared across cognitive substrates, but structured enough to determine recognizable linguistic patterns.

Definition 3.4 (Semantic-frame category)**.** The category Frame_Q has as objects semantic frames, including event frames, argument-role frames, force-dynamic frames, person and number frames, possession frames, spatial frames, attention frames, and other cognitive schemas relevant to linguistic expression. A morphism

$$r : F \to F'$$

is a frame refinement, abstraction, embedding, or transformation, graded by how much semantic structure it preserves, how much it abstracts away, and how costly it is to represent.

In this picture, a person hierarchy is not just a syntactic pattern: it is the externalized trace of a frame distinguishing speech-act participants from third parties. A case hierarchy is not just a list of morphological markings: it is the externalized trace of an argument-role frame. The frame category is where these cognitive patterns live before they become linguistic observations.

3.4 The linguistic category Lang_Q

Definition 3.5 (TUG linguistic object)**.** An object $L \in \mathsf{Lang}_Q$ is a typological TUG core

$$L = (M_L, \mathrm{Lab}_L, \perp_L, \Gamma_L, \mathcal{O}_{\mathrm{typ},L}, \mu_L)\,,$$

where

$$M_L = \mathrm{Lab}_L^* / \equiv_{\perp,\Gamma,\mathcal{O}_{\mathrm{typ}}}$$

is the trace quotient of visible grammatical actions by the maximal sound typological congruence. The interface

$$\mathcal{O}_{\mathrm{typ},L} = (\Sigma_{\mathrm{LF}}, \Sigma_{\mathrm{PF}}, \Sigma_{\mathrm{Morph}})$$

records the information needed for typological features. The valuation μ_L grades derivations, features, universals, or trace pairs.

Definition 3.6 (Linguistic morphism)**.** A morphism

$$h : L \to L'$$

in Lang_Q is an interface-preserving trace functor. It maps labels, traces, symmetries, independence relations, and typological predicates from L to L', preserving observations up to defect:

$$\begin{tikzcd}
M_L \arrow[r, "h"] \arrow[d, "\mathcal{O}_L"'] & M_{L'} \arrow[d, "\mathcal{O}_{L'}"] \\
\mathrm{Obs}_L \arrow[r, "\mathrm{Obs}(h)"'] & \mathrm{Obs}_{L'}\,.
\end{tikzcd}$$

The grade $\omega(h) \in Q$ records how much typological information, independence structure, symmetry structure, and weakness are preserved.

3.5 Pattern categories and subobjects

Universals are not usually whole languages or whole cognitive systems. They are patterns inside them. Categorically, patterns are represented by subobjects.

Definition 3.7 (Pattern as subobject)**.** For an object X in one of the categories above, a pattern in X is a subobject

$$P \hookrightarrow X.$$

The poset of subobjects of X is denoted $\mathrm{Sub}(X)$. A linguistic universal in a language core L is a subobject of $\mathrm{Obs}(L)$, or equivalently a predicate on L's typological core. A cognitive invariant in A is a subobject $I \hookrightarrow A$ stable under the relevant update endomorphisms.

For less mathematical readers: a subobject is the categorical version of a subset, property, or predicate. The universal "if a language has a dual, it has a plural" is not a language; it is a constraint on possible feature states. That constraint is represented as a subobject of the feature-state space.

4 The cognition–language correspondence

4.1 The representable case: a functor

In the most direct case, the cognition-language correspondence is a functor

$$\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$$

factoring through frames:

$$\begin{tikzcd} \mathsf{Cog}_Q \arrow[r, "\mathcal{K}"] \arrow[rr, bend right, "\Lambda"'] & \mathsf{Frame}_Q \arrow[r, "\mathcal{E}"] & \mathsf{Lang}_Q. \end{tikzcd}$$

Here $\mathcal{K}$ extracts from a cognitive system the semantic frames supported by its causal modules, and $\mathcal{E}$ externalizes frames into TUG trace structure. The composite $\Lambda = \mathcal{E} \circ \mathcal{K}$ maps a cognitive substrate to its linguistic externalization core.

Definition 4.1 (Cognition–language functor)**.** A cognition–language functor is a Q-weighted lax symmetric monoidal functor

$$\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$$

factoring as $\mathcal{E} \circ \mathcal{K}$, together with structural morphisms

$$\lambda_{A,B} : \Lambda(A) \otimes \Lambda(B) \to \Lambda(A \otimes B), \qquad \lambda_0 : I_{\mathsf{Lang}_Q} \to \Lambda(I_{\mathsf{Cog}_Q}),$$

whose grades record the degree to which linguistic externalization preserves cognitive factorization.

The adjective *lax* matters. Cognitive modules do not always externalize into perfectly independent linguistic modules. Sometimes externalization introduces couplings; sometimes it forgets distinctions; sometimes it pushes several cognitive factors through the same phonological or morphological channel. Lax monoidality expresses "preserves modular composition, but possibly with slack."

4.2 The non-representable case: a profunctor

A single cognitive system may support multiple linguistic externalizations, and a single linguistic pattern may be supported by multiple cognitive substrates. In that case a functor may be too deterministic. The more general object is a profunctor.

Definition 4.2 (Cognition–language profunctor)**.** A general cognition–language correspondence is a Q-enriched profunctor

$$\mathcal{P} : \mathsf{Cog}_Q^{op} \times \mathsf{Lang}_Q \to Q.$$

The value $\mathcal{P}(A, L) \in Q$ grades how well cognitive system A explains or realizes linguistic core L. The functorial case is recovered when

$$\mathcal{P}(A, L) = \mathsf{Lang}_Q(\Lambda A, L).$$

Conceptually, the profunctor says: instead of assigning one language core to one cognitive system, we assign a graded space of possible correspondences. This is often more realistic. Human cognition, typological data, and semantic frames are underdetermined in both directions.

4.3 Production and comprehension as an adjunction

The factorization $\Lambda = \mathcal{E} \circ \mathcal{K}$ separates frame extraction from linguistic externalization. The externalization functor $\mathcal{E} : \mathsf{Frame}_Q \to \mathsf{Lang}_Q$ should have an interpretation or comprehension partner

$$\mathcal{S} : \mathsf{Lang}_Q \to \mathsf{Frame}_Q.$$

The natural formal relation is an adjunction:

$$\mathcal{E} \dashv \mathcal{S}.$$

Definition 4.3 (Externalization–interpretation adjunction)**.** An externalization–interpretation adjunction consists of functors

$$\mathcal{E} : \mathsf{Frame}_Q \to \mathsf{Lang}_Q, \qquad \mathcal{S} : \mathsf{Lang}_Q \to \mathsf{Frame}_Q$$

and a natural isomorphism of hom-objects

$$\mathsf{Lang}_Q(\mathcal{E}F, L) \cong \mathsf{Frame}_Q(F, \mathcal{S}L).$$

The unit and counit are

$$\eta_F : F \to \mathcal{S}\mathcal{E}F, \qquad \epsilon_L : \mathcal{E}\mathcal{S}L \to L.$$

The adjunction means: producing a linguistic object from a semantic frame is equivalent, in the appropriate categorical sense, to interpreting that linguistic object as a semantic frame. The unit and counit express the two round-trips:

$$F \xrightarrow{\eta_F} \mathcal{S}\mathcal{E}F \qquad \mathcal{E}\mathcal{S}L \xrightarrow{\epsilon_L} L.$$

For readers with a linguistics background: this is a formal version of the production/comprehension duality. It does not say that production and comprehension are identical processes. It says their correctness conditions are paired by a universal relation.

4.4 Direct images and pullbacks of universals

Now we can state the core thesis precisely. Assume that the subobject fibrations over Cog_Q and Lang_Q admit pullback along Λ, and, on the relevant sublattices, direct image Σ_Λ left adjoint to pullback:

$$\Sigma_\Lambda \dashv \Lambda^*.$$

Thus for subobjects $I \hookrightarrow A$ and $U \hookrightarrow \Lambda A$,

$$\Sigma_\Lambda(I) \leq U \quad \iff \quad I \leq \Lambda^*(U).$$

Definition 4.4 (Direct image of a cognitive invariant)**.** Let $I \hookrightarrow A$ be a cognitive invariant. Its linguistic direct image is

$$\Sigma_\Lambda(I) \hookrightarrow \Lambda A.$$

When $\Sigma_\Lambda(I)$ is stable under the relevant linguistic dynamics, it is a linguistic universal induced by the cognitive invariant I.

Definition 4.5 (Pullback explanation of a linguistic universal)**.** Let $U \hookrightarrow L$ be a linguistic universal and let $\eta : \Lambda A \to L$ be a realization map. The cognitive pullback explanation of U along $\eta \circ \Lambda$ is the subobject $P_A \hookrightarrow A$ given by the pullback square

$$\begin{tikzcd} P_A \arrow[r] \arrow[d, hook] & U \arrow[d, hook] \\ A \arrow[r, "\eta\circ\Lambda"'] & L. \end{tikzcd}$$

A weakest pullback explanation is a maximal-weakness subobject among the pullback explanations compatible with the evidence.

This diagram is the precise form of the second boxed sentence from the abstract. A linguistic universal is a subobject U of the linguistic world. Pulling it back gives the cognitive structure required to produce it. Weakness then chooses the least overcommitted such structure.

4.5 Invariants transport to universals

Definition 4.6 (Invariant subobject)**.** Let $A \in \mathsf{Cog}_Q$. A subobject $I \hookrightarrow A$ is invariant under a family of updates $\{U_c\}_{c\in C}$ if for every c there exists an endomorphism $U_c|_I : I \to I$ making the square commute:

$$\begin{tikzcd} I \arrow[r, "U_c|_I"] \arrow[d, hook] & I \arrow[d, hook] \\ A \arrow[r, "U_c"'] & A. \end{tikzcd}$$

In the approximate/enriched case the square commutes up to a defect bounded in Q.

Theorem 4.7 (Direct image of invariants)**.** *Let $\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$ be a Q-weighted functor that maps cognitive updates U_c to linguistic actions $\Lambda(U_c)$ non-expansively with respect to invariance defects. Suppose the subobject fibration over Lang_Q admits direct image Σ_Λ along Λ on the relevant sublattice. If $I \hookrightarrow A$ is invariant under $\{U_c\}_{c\in C_A}$, then $\Sigma_\Lambda(I) \hookrightarrow \Lambda A$ is invariant under $\{\Lambda(U_c)\}$. In the approximate case, the linguistic invariance defect is bounded by the image under Λ's defect map of the cognitive invariance defect.*

Proof. In the strict case, invariance of I gives, for each c, an endomorphism $U_c|_I : I \to I$ with $\iota \circ U_c|_I = U_c \circ \iota$, where $\iota : I \hookrightarrow A$ is the inclusion. Applying Λ to this commuting square yields a commuting square in Lang_Q:

$$\Lambda(\iota) \circ \Lambda(U_c|_I) = \Lambda(U_c) \circ \Lambda(\iota).$$

The image $\Lambda(\iota) : \Lambda(I) \to \Lambda(A)$ is, in the relevant sublattice, the inclusion representing the direct image $\Sigma_\Lambda(I) \hookrightarrow \Lambda(A)$. The endomorphism $\Lambda(U_c|_I)$ restricts $\Lambda(U_c)$ to $\Sigma_\Lambda(I)$, making $\Sigma_\Lambda(I)$ stable under $\Lambda(U_c)$.

In the enriched case the inner square commutes only up to a defect $\delta_c^{\mathrm{cog}} \in Q$. Non-expansiveness of Λ on invariance defects means there is a quantale homomorphism φ such that the linguistic defect δ_c^{ling} satisfies

$$\delta_c^{\mathrm{ling}} \leq \varphi(\delta_c^{\mathrm{cog}}).$$

This is the stated bound. $\square$

This theorem is the precise form of the claim that linguistic universals are direct images of cognitive invariants. It is the building block for the closure-transport, factorization, and stability-rate transport theorems in Section **??**.

5 Weakest explanations and quantale selection

5.1 The explanation category

A linguistic universal generally has many possible explanations. Some are causal and sparse; others are correlation-based and redundant. We represent these possibilities categorically.

Definition 5.1 (Explanation category)**.** Let $U \hookrightarrow L$ be a linguistic universal. The category $\mathrm{Exp}(U)$ has as objects tuples

$$E = (A, F, L_E, \eta, \rho),$$

where

- $A \in \mathsf{Cog}_Q$ is a cognitive causal substrate;
- $F = \mathcal{K}(A) \in \mathsf{Frame}_Q$ is its extracted semantic-frame structure;
- $L_E = \mathcal{E}(F) \in \mathsf{Lang}_Q$ is its linguistic externalization;
- $\eta : L_E \to L$ is an interface-respecting realization map;
- $\rho : U \hookrightarrow L$ is witnessed by the image of L_E under η.

A morphism $E \to E'$ is a structure-preserving transformation of explanations commuting with the realization maps up to Q-defect.

For each explanation E choose a universe X_E of relevant states, frames, traces, or feature configurations. Let

$$H_E \subseteq X_E \times X_E$$

be the indistinction relation induced by E: $(x, y) \in H_E$ means that explanation E does not distinguish x and y. Its base weakness is

$$w(H_E) = \bigvee_{(x,y) \in H_E} \mu_E(x) \otimes \mu_E(y).$$

Definition 5.2 (Explanation weakness)**.** The full weakness grade of an explanation E is the product-quantale value

$$W(E) = \big(w(H_E), \mathrm{Fit}(E), \mathrm{Cost}(E), \mathrm{Conf}(E), \mathrm{DC}(E)\big) \in Q_{\mathrm{ind}} \times Q_{\mathrm{fit}} \times Q^{op}_{\mathrm{cost}} \times Q^{op}_{\mathrm{conf}} \times Q^{op}_{\mathrm{dc}}.$$

A weakest explanation of U is any

$$E^* \in \arg\max_{E \in \mathrm{Exp}(U)} W(E).$$

Because the cost, confusion, and double-counting components are ordered oppositely, maximizing W means maximizing indistinction and fit while minimizing cost, confusion, and double-counting.

5.2 Relation to TyLAA weakness

TyLAA selects the weakest red operational explanation consistent with a blue type/proof interface. The framework here selects the weakest cognitive causal explanation consistent with a linguistic universal. These are not two unrelated Occam principles; they are two layers of the same quantale-enriched selection process, and the next definition and theorem make their composition precise.

Definition 5.3 (Weakness-preserving correspondence)**.** Let W_C be a cognitive weakness measure on indistinction relations over a cognitive universe, and let W_L be a linguistic/TUG weakness measure on indistinction relations over a linguistic universe. A cognition-language functor Λ is *weakness-preserving* if it acts on indistinction relations H by sending each pair $(x, y) \in H$ to a pair $(\Lambda x, \Lambda y) \in \Lambda H$, and there exists a quantale homomorphism

$$\varphi : Q_C \to Q_L$$

satisfying both of the following:

1. (Lower bound) For every indistinction relation H,

$$W_L(\Lambda H) \geq \varphi(W_C(H)),$$

   with equality in the strict case.

2. (Monotonicity on relations) Whenever $H \subseteq H'$ as indistinction relations, $\Lambda H \subseteq \Lambda H'$.

The first clause says that Λ does not destroy the cognitive weakness of an indistinction relation when it pushes it forward. The second clause says that Λ is order-preserving on the lattice of indistinction relations: enlarging the cognitive equivalences only enlarges the linguistic ones. The second clause is automatic in the strict case where ΛH is defined pointwise.

Theorem 5.4 (Composition of TyLAA and cognitive weakness)**.** *Assume Λ is weakness-preserving in the sense of Definition* **??***, and that W_L is itself $\subseteq$-monotone (a property which holds automatically for the join-style measure $w(H) = \bigvee_{(x,y)\in H} \mu(x) \otimes \mu(y)$). Let H^*_C be the largest sound cognitive congruence compatible with the semantic-frame interface, in the sense that every other compatible sound cognitive congruence H satisfies $H \subseteq H^*_C$. Then $\Lambda(H^*_C)$ is maximally weak among linguistic congruences in the image of Λ. If Λ is essentially surjective on sound linguistic congruences, then the TyLAA weakest-red congruence and the image of the weakest cognitive congruence coincide on the image.*

Proof. Let K be any sound linguistic congruence in the image of Λ, say $K = \Lambda(H)$ for a sound cognitive congruence H compatible with the interface. By maximality of H_C^*, we have $H \subseteq H_C^*$. Applying clause (2) of Definition **??**,

$$\Lambda H \subseteq \Lambda H_C^*.$$

Applying $\subseteq$-monotonicity of W_L,

$$W_L(K) = W_L(\Lambda H) \leq W_L(\Lambda H_C^*).$$

Hence ΛH_C^* is maximally weak among linguistic congruences in the image of Λ. If Λ is essentially surjective on sound linguistic congruences, every sound linguistic congruence is isomorphic to one in the image, so the same maximality holds up to isomorphism. $\square$

This theorem is the formal version of the intuition that TyLAA weakness composes with causal-coding weakness rather than sitting beside it as an unrelated principle. The cognitive side decides which derivations should be regarded as the same; the linguistic side inherits that judgment along Λ, and the monotonicity clause is what guarantees that the two choices align.

5.3 Why causal explanations beat double-counting explanations

The quantale order makes precise why causal explanations are preferred over redundant correlation stories.

Proposition 5.5 (Causal explanations dominate redundant explanations)**.** *Let E_{causal} and E_{corr} be two explanations of the same linguistic universal U. Suppose they have equal or comparable fit and that*

$$w(H_{E_{\text{causal}}}) \geq w(H_{E_{\text{corr}}}),$$

$$\mathrm{Cost}(E_{\text{causal}}) \leq \mathrm{Cost}(E_{\text{corr}}), \qquad \mathrm{Conf}(E_{\text{causal}}) \leq \mathrm{Conf}(E_{\text{corr}}), \qquad \mathrm{DC}(E_{\text{causal}}) < \mathrm{DC}(E_{\text{corr}}).$$

Then

$$W(E_{\text{causal}}) > W(E_{\text{corr}})$$

in the product quantale order, provided at least one of the displayed inequalities is strict in a non-ignored component.

Proof. This is immediate from the componentwise product order and the use of opposite order in the cost-like components. Equal or better fit, greater or equal indistinction, lower or equal cost/confusion, and strictly lower double-counting imply strict dominance. $\square$

The proposition is simple, but it is important. A causal explanation wins not because "causal" is a magic word, but because a genuine mediator lets the theory represent one dependency once. A correlation-only explanation often has to represent the same dependency through many overlapping pairwise links, increasing double-counting and confusion.

6 Main theorem: monoidal transport of causal modularity

The main theorem of the paper says, roughly: cognitive modularity is mapped by Λ to TUG modularity, and the failure of cognitive modularity is bounded by, and transported into, the failure of TUG modularity. This is the precise version of the informal claim that the TUG factorization theorem and the causal-continual-learning commutator theorem are two faces of one structural fact.

6.1 Commutators and swap defects

To state the theorem we need a way to measure non-commutativity on both the cognitive and the linguistic side. Let $A \in \mathsf{Cog}_Q$ have context updates $U_c, U_d : A \to A$. Their cognitive commutator defect is

$$\mathrm{Comm}_C(c,d) := d_C(U_c \circ U_d, U_d \circ U_c),$$

where d_C is the (quantale-valued) defect metric on cognitive endomorphisms. Let $L = \Lambda A$. The images $\Lambda(U_c), \Lambda(U_d)$ are linguistic actions, i.e. trace transformations on the TUG core. Their linguistic swap defect is

$$\mathrm{Comm}_L(c,d) := d_L(\Lambda(U_c) \circ \Lambda(U_d), \Lambda(U_d) \circ \Lambda(U_c)).$$

The functor Λ is *non-expansive with respect to commutator defects* if there is a quantale homomorphism $\varphi : Q_C \to Q_L$ such that

$$\mathrm{Comm}_L(c,d) \le \varphi(\mathrm{Comm}_C(c,d))$$

for all pairs of contexts c, d. Read in the defect order, this inequality says that linguistic non-commutativity is bounded by the image of cognitive non-commutativity. Equivalently, read in the weakness order, the linguistic commutativity grade is at least the image of the cognitive commutativity grade: closer-to-commuting cognitive updates produce closer-to-commuting linguistic actions.

6.2 The transport theorem

Theorem 6.1 (Enriched monoidal transport of causal modularity into TUG modularity)**.** *Let*

$$\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$$

be a Q-weighted lax symmetric monoidal functor, non-expansive with respect to commutator defects. Let

$$A \cong A_1 \otimes \cdots \otimes A_n$$

be a factored cognitive object with context-local update endomorphisms U_c supported on $S_c \subseteq \{1, \ldots, n\}$. Then:

1. *If $S_c \cap S_d = \emptyset$, then U_c and U_d commute up to the canonical monoidal symmetry, and their linguistic images are swap-equivalent in the TUG trace quotient.*
2. *If $\mathrm{Comm}_C(c,d) \le \varepsilon$, then*
$$\mathrm{Comm}_L(c,d) \le \varphi(\varepsilon).$$
3. *If $A = A_I \otimes A_J$ and Λ is strong monoidal on this split, then*
$$\Lambda(A) \cong \Lambda(A_I) \otimes \Lambda(A_J).$$
4. *Consequently, any nontrivial universal implication between predicates living purely on $\Lambda(A_I)$ and $\Lambda(A_J)$ requires either failure of strong monoidality on the split or a mediator object connecting the two factors.*

Proof. For (1), context-locality says that U_c acts as the identity outside S_c, and U_d acts as the identity outside S_d. If the supports are disjoint, the two endomorphisms act on different tensor factors. In a symmetric monoidal category, endomorphisms on disjoint tensor factors commute up to the canonical braiding and coherence isomorphisms. Applying Λ, functoriality sends this commutation to a commutation of linguistic trace actions. In the TUG quotient, adjacent independent actions are identified by the independence congruence, so the linguistic images are swap-equivalent.

For (2), non-expansiveness is exactly the displayed inequality.

For (3), strong monoidality on the split gives an isomorphism $\Lambda(A_I \otimes A_J) \cong \Lambda(A_I) \otimes \Lambda(A_J)$.

For (4), predicates on separate factors vary independently in a product. If an implication $\alpha(x) \Rightarrow \beta(y)$ holds for all $(x, y) \in X \times Y$, then either no x satisfies α, making the implication vacuous, or every y satisfies β, making it trivial. A nontrivial cross-factor implication therefore requires some structure beyond the product: either monoidality fails to be strong, or a mediator introduces a coupling. □

The theorem says that the TUG factorization theorem and the causal-continual-learning commutator theorem are the same theorem transported through Λ. The TUG theorem is the linguistic side of modularity; the CCL theorem is the cognitive-learning side.

6.3 The core commutative diagram

The transport content of the theorem can be summarized by the following diagram in which the cognitive commutator on the left is sent by Λ to the linguistic swap defect on the right:

$$\begin{tikzcd} A \arrow[r, shift left, "U_c U_d"] \arrow[r, shift right, "U_d U_c"'] \arrow[d, "\Lambda"'] & A \arrow[d, "\Lambda"] \\ \Lambda A \arrow[r, shift left, "\Lambda(U_c)\Lambda(U_d)"] \arrow[r, shift right, "\Lambda(U_d)\Lambda(U_c)"'] & \Lambda A \end{tikzcd}$$

On the upper edge, the two arrows represent the two orders of cognitive update; their gap is bounded by $\mathrm{Comm}_C(c, d)$. On the lower edge, the two arrows represent the two orders of linguistic action; their gap is bounded by $\mathrm{Comm}_L(c, d) \le \varphi(\mathrm{Comm}_C(c, d))$. The vertical Λ arrows transport the upper bound to the lower bound. When the supports of U_c and U_d are disjoint, both pairs of arrows coincide up to the appropriate canonical isomorphism, and in particular the linguistic pair is identified by the TUG independence congruence.

7 Mediators as residuals

7.1 Why broad universals need mediators

Broad universals are claims that try to connect features from different grammatical modules: a syntactic feature with a morphological one, a phonological externalization choice with a semantic-role feature, or similar. By Theorem **??**, if the cognitive substrate factors and Λ is strong monoidal on the split, no nontrivial implication can cross the boundary in the linguistic image. Therefore any *genuine* broad universal must involve some structure beyond the strict product: either failure of strong monoidality on the split, or an explicit mediator that couples the two factors.

A mediator, in this framework, is not just an informal third variable. Categorically, it is the weakest context that makes a cross-module implication true. The next subsection makes "weakest" precise via residuation in a behavior quantale.

7.2 Residuals in a behavior quantale

Let M be a monoid of traces or behaviors, and let $\mathcal{P}(M)$ be its powerset quantale. Sequential composition of behaviors is

$$A \cdot B := \{ab : a \in A,\ b \in B\}.$$

The right residual is

$$A \Rightarrow C := \{m \in M : A \cdot \{m\} \subseteq C\}.$$

It satisfies the residuation law

$$A \cdot B \subseteq C \quad \Longleftrightarrow \quad B \subseteq (A \Rightarrow C).$$

Thus $A \Rightarrow C$ is the largest, hence weakest, behavior B such that $A \cdot B \subseteq C$.

Theorem 7.1 (Mediator as weakest residual context)**.** *Let P_A and P_B be behavior predicates associated with two factored modules. A mediator M makes the cross-module implication from P_A to P_B valid precisely when*

$$P_A \cdot M \subseteq P_B.$$

The weakest such mediator is the residual

$$M^* = P_A \Rightarrow P_B.$$

Proof. By the residuation law,

$$P_A \cdot M \subseteq P_B \quad \Longleftrightarrow \quad M \subseteq P_A \Rightarrow P_B.$$

Therefore $P_A \Rightarrow P_B$ is the largest behavior set satisfying the condition. Since larger behavior sets are weaker, it is the weakest mediator. □

Linguistically, this is the formal version of saying that case marking, argument-structure resolution, agreement morphology, or information-structure marking can mediate between modules. The mediator is not an arbitrary extra assumption; it is the weakest context needed to make the cross-module implication nontrivial.

7.3 A mediator diagram

The mediator breaks the product by introducing a span or cospan:

$$\begin{tikzcd} & M \arrow[dl] \arrow[dr] & \\ A & & B \end{tikzcd} \qquad \text{or} \qquad \begin{tikzcd} A \arrow[dr] & & B \arrow[dl] \\ & M & \end{tikzcd}$$

The first diagram reads M as a shared source or resource; the second reads M as a shared target or interface. Case marking is often closer to the second: syntactic and semantic-role information both feed into a morphological/externalization mediator.

8 The five correspondences as categorical transport results

We now formulate the five correspondences linking the TUG conclusions to the causal-coding architectural features in the categorical language developed above. The status of the five claims is not uniform. The factorization correspondence (Correspondence 2) is an identity theorem: it is exactly Theorem **??** and Theorem **??**, and follows from the general structure of Λ without further assumptions. The remaining four are *conditional* transport results: each says that, under explicit and stated assumptions about the structure of Λ, cognitive structure maps to a specific linguistic structure. Where assumptions remain hypothetical, we flag them as such; the empirical research programme of Section **??** is largely the task of testing these assumptions.

8.1 Correspondence 1: hierarchical universals as closure transport

Conceptual version

A hierarchy such as person, number, case, or accessibility says: if a system has a more marked feature, it must have the less marked features below it. Mathematically, this is a closure system. Cognitively, the hypothesis is that the relevant hierarchy is carried by stable semantic-frame substrate: a hard-kernel or inner-shell frame whose structure is defended across contexts. Linguistically, the hierarchy is the direct image of that cognitive closure structure.

Mathematical formulation

Let P_C be a finite poset of cognitive-frame features and let

$$\mathrm{cl}_C : \mathcal{P}(P_C) \to \mathcal{P}(P_C)$$

be a closure operator generated by cognitive implications. Let P_L be the corresponding linguistic feature poset. Suppose the cognition-language correspondence induces a Galois connection

$$\Sigma_\Lambda : \mathcal{P}(P_C) \rightleftarrows \mathcal{P}(P_L) : \Lambda^*.$$

Theorem 8.1 (Closure transport)**.** *Assume that, on the relevant feature sublattice,*

$$\Sigma_\Lambda \Lambda^* = \mathrm{id}$$

and that $\Lambda^*\Sigma_\Lambda$ sends cl_C-closed sets to cl_C-closed sets. Define

$$\mathrm{cl}_L(S) := \Sigma_\Lambda\big(\mathrm{cl}_C(\Lambda^* S)\big).$$

Then cl_L is a closure operator on $\mathcal{P}(P_L)$. Its fixed points are precisely the linguistic feature states whose cognitive pullbacks are cl_C-closed.

Proof. Extensivity follows from the unit of the Galois connection ($S \subseteq \Sigma_\Lambda \Lambda^* S$, here an equality) combined with the extensivity of cl_C:

$$S = \Sigma_\Lambda \Lambda^* S \subseteq \Sigma_\Lambda \,\mathrm{cl}_C\, \Lambda^* S = \mathrm{cl}_L(S).$$

Monotonicity follows from monotonicity of Λ^*, cl_C, and Σ_Λ and from the fact that they compose monotonically.

For idempotence, expand:

$$\mathrm{cl}_L(\mathrm{cl}_L(S)) = \Sigma_\Lambda \,\mathrm{cl}_C\, \Lambda^*\big(\Sigma_\Lambda \,\mathrm{cl}_C\, \Lambda^* S\big) = \Sigma_\Lambda \,\mathrm{cl}_C\big(\Lambda^* \Sigma_\Lambda (\mathrm{cl}_C\, \Lambda^* S)\big).$$

By assumption, $\Lambda^*\Sigma_\Lambda$ sends cl_C-closed sets to cl_C-closed sets, and $\mathrm{cl}_C \Lambda^* S$ is cl_C-closed. Hence $\Lambda^*\Sigma_\Lambda(\mathrm{cl}_C \Lambda^* S)$ is cl_C-closed, so applying cl_C again leaves it fixed:

$$\mathrm{cl}_C\big(\Lambda^*\Sigma_\Lambda(\mathrm{cl}_C \Lambda^* S)\big) = \Lambda^*\Sigma_\Lambda(\mathrm{cl}_C \Lambda^* S).$$

Applying Σ_Λ and using $\Sigma_\Lambda\Lambda^* = \mathrm{id}$ on the relevant sublattice yields

$$\mathrm{cl}_L(\mathrm{cl}_L(S)) = \Sigma_\Lambda\Lambda^*\Sigma_\Lambda(\mathrm{cl}_C \Lambda^* S) = \Sigma_\Lambda(\mathrm{cl}_C \Lambda^* S) = \mathrm{cl}_L(S),$$

which is idempotence. The fixed-point characterization is then immediate from the construction. $\square$

This theorem turns hierarchical linguistic universals into the direct image of cognitive closure. The hard-kernel story is a proposed source of cl_C; the TUG hierarchy is its externalized cl_L.

8.2 Correspondence 2: cross-module universals and factorization

This is Theorem **??** and Theorem **??**. If the cognitive substrate factors and Λ is strong monoidal on the split, the linguistic core factors. A broad cross-module universal is then impossible except in vacuous/trivial cases. Therefore any surviving broad universal points to a mediator, and the weakest mediator is a residual.

The conceptual message is strong: broad universals are rare not because linguists failed to formulate them cleverly, but because a modular continual learner should not produce many global cross-module laws. Where broad universals survive, they are evidence for genuine bridge structure.

8.3 Correspondence 3: narrow word order as sparse-energy pushforward

Conceptual version

Word order is not one global parameter. It is a network of local externalization choices with coherence pressures among them. Cognitively, these pressures arise from sparse routing among semantic and production modules. Linguistically, they appear as a sparse signed interaction graph among word-order features.

Mathematical formulation

Let S_C be a finite or compact cognitive controller state space, and let

$$\pi : S_C \to X_L$$

map controller states to linguistic word-order configurations. Suppose the cognitive controller has a sparse signed energy

$$E_C(s) = \frac{1}{2}\sum_{i<j} a_{ij}(1 - t_{ij}s_i s_j),$$

where $a_{ij} \geq 0$ and $t_{ij} \in \{\pm 1\}$. Define the linguistic pushforward energy by Gibbs marginalization:

$$e^{-\beta E_L(x)} := \sum_{\pi(s)=x} e^{-\beta E_C(s)}.$$

Proposition 8.2 (Sparse-energy pushforward)**.** *Assume π is local in the sense that each linguistic coordinate depends on a bounded number of cognitive coordinates, and assume the higher-order cumulants induced by marginalizing over fibers of π are bounded by ε. Then E_L is approximated by a sparse signed pairwise energy*

$$E_L(x) = \frac{1}{2}\sum_{i<j} w_{ij}(1 - s_{ij}x_i x_j) + R(x), \qquad \|R\| \leq O(\varepsilon).$$

Proof sketch. Gibbs pushforward expresses E_L as the negative logarithm of a marginal partition sum. The logarithm admits a cumulant expansion over the linguistic coordinates. Locality of π and sparsity of E_C ensure that low-order terms dominate and that induced interactions have bounded neighborhood size. The assumption on higher-order cumulants bounds the residual R. The pairwise coefficients w_{ij} and signs s_{ij} are the second-order cumulants of the pushed-forward energy. □

Thus narrow word-order universals are the externalization-side pushforward of sparse controller coherence.

8.4 Correspondence 4: graded stability as rate transport

Conceptual version

Universals are not simply true or false. Some are hard-kernel stable; some are middle-shell tendencies; some are outer-shell residue; some are spurious. The linguistic stability coefficient proposed in [**?**] should be the image of a cognitive stability coefficient under Λ.

Mathematical formulation

Let a cognitive feature p have two-state transition rates

$$0 \xrightarrow{\alpha_C} 1, \qquad 1 \xrightarrow{\beta_C} 0.$$

Let $U = \Lambda(p)$ be the corresponding linguistic feature or universal, with rates α_L, β_L. Define

$$\sigma_C(p) = \log\frac{\alpha_C}{\beta_C}, \qquad \sigma_L(U) = \log\frac{\alpha_L}{\beta_L}.$$

Theorem 8.3 (Stability-rate transport)**.** *Assume Λ distorts transition rates by at most a multiplicative factor $e^{\pm\varepsilon}$:*

$$e^{-\varepsilon} \leq \frac{\alpha_L}{\alpha_C} \leq e^{\varepsilon}, \qquad e^{-\varepsilon} \leq \frac{\beta_L}{\beta_C} \leq e^{\varepsilon}.$$

Then

$$|\sigma_L(U) - \sigma_C(p)| \leq 2\varepsilon.$$

Proof. Write $\alpha_L = \alpha_C e^{r_\alpha}$ and $\beta_L = \beta_C e^{r_\beta}$, with $|r_\alpha|, |r_\beta| \leq \varepsilon$. Then

$$\sigma_L(U) - \sigma_C(p) = \log\frac{\alpha_C e^{r_\alpha}}{\beta_C e^{r_\beta}} - \log\frac{\alpha_C}{\beta_C} = r_\alpha - r_\beta,$$

so the absolute value is at most 2ε. □

This theorem links linguistic stability to cognitive stability. A hard-kernel cognitive invariant, for which β_C is extremely small, maps to a high-stability linguistic universal unless Λ dramatically distorts rates.

8.5 Correspondence 5: context-indexed laws as gluing

Conceptual version

Many universals are local: they hold in subordinate clauses, or in definite object contexts, or in animate-object contexts, or inside a particular phrase type. The cognitive analogue is controller selection: different contexts activate different module supports. The mathematical structure is an indexed family of local closure systems glued over context overlaps.

Mathematical formulation

Let Ctx be a category of contexts. A context-indexed grammar/cognition system is a functor

$$\mathcal{F} : \mathsf{Ctx}^{op} \to \mathsf{Poset}$$

assigning to each context c a feature poset P_c, and to each context refinement $r : d \to c$ a restriction map $r^* : P_c \to P_d$. Suppose each context has a local closure operator

$$\mathrm{cl}_c : \mathcal{P}(P_c) \to \mathcal{P}(P_c).$$

Theorem 8.4 (Exact gluing of local closure laws)**.** *Assume that for every overlap/refinement $r : d \to c$, local closure commutes with restriction:*

$$r^* \,\mathrm{cl}_c(S) = \mathrm{cl}_d(r^* S).$$

Then the family $\{\mathrm{cl}_c\}$ glues to a global closure operator on compatible sections of $\mathcal{F}$.

Proof. A compatible section is a family $(S_c)_c$ satisfying $r^* S_c = S_d$ for every refinement $r : d \to c$. Define

$$\mathrm{cl}((S_c)_c) := (\mathrm{cl}_c(S_c))_c.$$

The commutation assumption ensures compatibility of the resulting family. Extensivity, monotonicity, and idempotence hold componentwise because each cl_c is a closure operator. □

Proposition 8.5 (Approximate gluing defect)**.** *If the commutation equation fails by defects δ_r, then the global closure defect is bounded by the join or sum of the overlap defects, depending on the chosen quantale:*

$$\delta_{\mathrm{glue}} \leq \bigvee_{r \in \mathrm{Mor}(\mathsf{Ctx})} \delta_r.$$

When the defects are induced by cognitive context-update commutators, this is bounded by the corresponding confusion-pair costs from the approximate CCL theorem.

This theorem formalizes why broad aggregation can fail. A local law may be perfectly valid in every context, yet disappear when the context index is forgotten. Forgetting context is a functor that may fail to preserve closure.

9 Semantic frames and GTGI-style cognitive mappings

9.1 Why the frame layer is essential

The functor $\Lambda : \mathsf{Cog}_Q \to \mathsf{Lang}_Q$ factors through Frame_Q because cognitive modules do not usually map directly to typological features. A causal module may support a cluster of person, agency, attention, event, or action schemas. A linguistic feature externalizes only part of that schema, often through an interface that mixes semantics, phonology, and morphology.

The patternist perspective in [?] treats mind as a system of patterns, including patterns in perception, action, memory, inference, attention, and self-modeling. In this paper, semantic frames are the category-theoretic carriers of those patterns insofar as they are relevant to language. They are the objects that make mappings from cognition to language compositional.

9.2 Frame morphisms

A frame morphism

$$r : F \to F'$$

can represent:

- abstraction: a detailed event frame maps to a coarser event frame;
- refinement: a general argument-role frame maps to a more specific case frame;
- embedding: a person frame embeds into a speech-act frame;
- metaphor or transfer: a spatial frame maps to an abstract relational frame;
- externalization preparation: a semantic frame maps to a frame enriched with morphosyntactic slots.

The grade $\omega(r)$ records how much frame structure is preserved and how much cost or ambiguity is introduced.

9.3 Natural transformations among cognitive modalities

The GTGI-style view emphasizes that language is not isolated. It interacts with perception, action, memory, inference, and attention. Categorically, these can be represented as functors into the frame category:

$$\begin{tikzcd}
\mathsf{Perception} \arrow[dr, "P"'] & & \mathsf{Action} \arrow[dl, "A"] \\
& \mathsf{Frame}_Q & \\
\mathsf{Memory} \arrow[ur, "M"] & & \mathsf{Lang}_Q \arrow[ul, "\mathcal{S}"']
\end{tikzcd}$$

Natural transformations between these functors express cognitive synergies: ways in which perception-guided frames, action-guided frames, memory-guided frames, and language-guided frames systematically correspond. The language-to-cognition direction is $\mathcal{S} : \mathsf{Lang}_Q \to \mathsf{Frame}_Q$; the cognition-to-language direction is $\mathcal{E} : \mathsf{Frame}_Q \to \mathsf{Lang}_Q$. Their adjunction formalizes the production/comprehension pairing.

9.4 Sapir–Whorf and its dual

The Sapir–Whorf question asks how language shapes cognition. In this framework that is a question about the composite

$$\mathsf{Lang}_Q \xrightarrow{\mathcal{S}} \mathsf{Frame}_Q \xleftarrow{\mathcal{K}} \mathsf{Cog}_Q.$$

The dual question, central here, asks how cognition shapes language:

$$\mathsf{Cog}_Q \xrightarrow{\mathcal{K}} \mathsf{Frame}_Q \xrightarrow{\mathcal{E}} \mathsf{Lang}_Q.$$

The typological universals record is evidence about the second direction. If a linguistic universal is stable across languages after genealogical and areal correction, then its pullback along Λ is evidence for a stable cognitive frame or invariant. The weakness principle says we should infer the weakest such invariant.

9.5 Three-tier classification of semantic frames

The stability coefficient σ suggests a three-tier classification:

Kernel frames. Frames whose linguistic images have very high stability. Candidate examples include basic person distinctions, basic number distinctions, argument-role accessibility, and elementary event/agent/patient structures.

Consolidated shell frames. Frames whose linguistic images are robust but exceptionful. Candidate examples include many narrow word-order coherence tendencies and moderately stable externalization preferences.

Outer-shell residue. Frames or feature couplings whose apparent universality disappears under correction or holds only in narrow contexts. These are culturally, historically, or locally learned regularities rather than protected substrate.

This classification is not merely linguistic. It is a hypothesis about cognitive architecture, inferred through linguistic direct images and pullbacks.

10 Semantic growth universals: word-meaning evolution as a dynamic cognition–language correspondence

The preceding sections have treated linguistic universals primarily as constraints on grammatical externalization: closure laws in paradigms, factorization and mediation laws between grammatical modules, low-frustration coherence laws for word order, graded stability laws for typological states, and context-indexed local laws. Recent empirical work on the statistical structure of word-meaning evolution [?] suggests that this same picture should be extended from grammar space to semantic-growth space.

The basic point is simple. Morphosyntactic universals tell us something about which linguistic forms are stable. Semantic-evolution universals tell us something about how meaning space itself grows. In the correspondence developed above, the relevant chain is

$$\mathsf{Cog}_Q \xrightarrow{\mathcal{K}} \mathsf{Frame}_Q \xrightarrow{\mathcal{E}} \mathsf{Lang}_Q.$$

The semantic-growth picture inserts a dynamic lexical-semantic layer between frames and observed linguistic form, and an observation functor sending it to a category of empirical embeddings:

$$\mathsf{CogDyn}_Q \xrightarrow{\mathcal{K}_\bullet} \mathsf{FrameDyn}_Q \xrightarrow{\mathcal{L}_\bullet} \mathsf{LexSemDyn}_Q \xrightarrow{\mathcal{O}} \mathsf{Emb}_Q.$$

Here $\mathsf{LexSemDyn}_Q$ is a category of evolving lexical-semantic spaces, and Emb_Q is a category of empirical embedding spaces equipped with quantale-valued distortion. Conceptually, $\mathcal{K}_\bullet$ extracts the active cognitive-frame dynamics, $\mathcal{L}_\bullet$ lexicalizes those frame dynamics as evolving word/concept regions, and $\mathcal{O}$ observes the result through a corpus-derived embedding.

Thus the slogan

linguistic universals are direct images of cognitive invariants

should be supplemented by the dynamic slogan

semantic-evolution universals are invariants of the frame-to-lexicon growth process,

and conversely

their cognitive pullbacks are attention-weighted, hierarchical,
context-local, bursty mechanisms of concept formation.

10.1 Empirical motivation: four regularities of semantic evolution

Guo et al. report four regularities in empirical embedding spaces built from English and 21 additional languages [**?**]:

1. *frequency assortativity*, where high-popularity entities tend to occur near other high-popularity entities;
2. characteristic *clustering velocity profiles*, reflecting hierarchical aggregation of cluster structure;
3. *persistent local temporal dynamics*, in which newly created entities appear disproportionately near other recently created entities;
4. *Taylor's law*, in which the variance of new-entity counts scales as a power of the mean across regions and time windows.

A class of directed preferential placement (DPP) generative models reproduces this profile in synthetic embeddings, while simpler uniform or purely preferential models fail to capture it.

For our purposes, the important point is not the specific embedding algorithm. It is that the regularities live in an inferred semantic or epistemic space: a space whose points are words, concepts, or knowledge entities, whose metric approximates semantic relatedness, whose valuations encode popularity or frequency, and whose time labels record lexical birth or historical emergence. Where the morphosyntactic universal program asks *which regions of grammar space are stable*, semantic-evolution work asks *by what laws lexical-semantic space expands*. The corresponding universals are not predicates of the form $X \Rightarrow Y$; they are invariants of stochastic growth processes.

10.2 Lexical-semantic growth objects

We formalize this by defining a category of evolving lexical-semantic spaces.

Definition 10.1 (Lexical-semantic growth object)**.** A lexical-semantic growth object is a time-indexed family

$$L_\bullet = \{L_t\}_{t \in T}, \qquad L_t = (X_t, d_t, \mu_t, \tau_t),$$

where X_t is a finite or locally finite set of lexical-semantic entities, $d_t : X_t \times X_t \to [0, \infty]$ is a semantic distance or dissimilarity, $\mu_t : X_t \to [0, \infty)$ is a valuation such as frequency, popularity, or cultural salience, and $\tau_t : X_t \to T$ records the first observed birth time of each entity. We require $X_t \subseteq X_{t'}$ for $t \le t'$, except when one explicitly models lexical death or deletion.

A morphism $f : L_\bullet \to L'_\bullet$ is a family of maps $f_t : X_t \to X'_{\alpha(t)}$, for a monotone time map $\alpha : T \to T'$, preserving distances, valuations, and birth order up to a chosen quantale-valued distortion. Thus f may compare languages, embedding algorithms, corpora, or historical periods without requiring exact identity of the underlying spaces.

Definition 10.2 (Semantic growth operator)**.** A semantic growth operator is a stochastic kernel

$$B_t : L_t \rightsquigarrow L_{t+1}$$

that adds new entities, changes valuations, or changes local distances. In the simplest birth-only case, $X_{t+1} = X_t \cup N_t$, where N_t is the set of entities born during the interval $[t, t+1]$.

A semantic-evolution universal is then an invariant, or approximately invariant, of the family $B_\bullet$.

Definition 10.3 (Semantic growth universal)**.** Let $\mathcal{T}(\mathsf{LexSemDyn}_Q)$ denote the category of trajectories in $\mathsf{LexSemDyn}_Q$. A semantic growth universal is a subobject $U \hookrightarrow \mathcal{T}(\mathsf{LexSemDyn}_Q)$ or a Q-valued predicate on trajectories that is stable under admissible morphisms of lexical-semantic growth objects. Equivalently, it is a property of semantic growth that survives changes of corpus, embedding method, and language, up to the allowed Q-defect.

Informally, a semantic growth universal is a stable law of how meaning space expands, not merely a stable co-occurrence relation among already-existing words.

10.3 The commuting diagram of cognitive and lexical growth

The intended correspondence is that active cognitive supports generate active semantic-frame regions, which in turn generate new lexical material. For a context c, let U_c be the cognitive update acting on a support S_c, let G_c be the induced frame update, and let B_c be the corresponding lexical birth or semantic-growth operator. The desired naturality condition is the approximate commutativity of the following diagram:

$$\begin{tikzcd}
C_t \arrow[r, "U_c"] \arrow[d, "\mathcal{K}_t"'] & C_{t+1} \arrow[d, "\mathcal{K}_{t+1}"] \\
F_t \arrow[r, "G_c"] \arrow[d, "\mathcal{L}_t"'] & F_{t+1} \arrow[d, "\mathcal{L}_{t+1}"] \\
X_t \arrow[r, "B_c"'] & X_{t+1}.
\end{tikzcd}$$

Updating the cognitive substrate and then externalizing it as lexical-semantic space should approximately agree with first extracting the semantic frame and then allowing lexical growth inside the corresponding region. The square need not commute strictly; it should commute up to a quantale-valued defect measuring semantic distortion, historical noise, polysemy, corpus limitations, and embedding error. In this form, empirical semantic-growth laws become observable shadows of cognitive growth laws.

10.4 Frequency assortativity as an attention-field universal

The first regularity is frequency assortativity: high-popularity entities tend to occur near other high-popularity entities in semantic space. Let $N_k(x)$ be the set of k nearest neighbors of x under d_t, and write

$$\bar{\mu}_{t,k}(x) = \frac{1}{k} \sum_{y \in N_k(x)} \mu_t(y).$$

A simple version of the assortativity statistic is

$$\rho_{\mu,k}(t) = \operatorname{corr}\left(\log \mu_t(x), \log \bar{\mu}_{t,k}(x)\right).$$

Frequency assortativity is the claim that, across the relevant languages, corpora, and embedding choices, $\rho_{\mu,k}(t) > 0$ for the relevant range of k and t, modulo the chosen statistical uncertainty.

The cognitive pullback of this fact is an attention-field universal. Human and cultural attention is not uniformly distributed over concept space. Highly salient concepts do not appear as isolated high-frequency atoms; they live in neighborhoods of other salient concepts. In the causal-coding picture, these neighborhoods are the lexical projection of heavily reused cognitive supports or semantic frames. High-frequency semantic regions are therefore good candidates for the lexical shadow of hard-kernel or middle-shell cognitive substrate: conceptual regions that many contexts touch, and which therefore accumulate dense high-frequency lexical neighborhoods.

10.5 Clustering velocity as a semantic hierarchy universal

The second regularity concerns clustering velocity profiles. Empirical semantic spaces appear to have characteristic multiresolution cluster-merging behavior, naturally expressed using a filtration. For each scale parameter ϵ, define a graph or simplicial complex

$$\mathcal{R}_\epsilon(X_t)$$

by joining points whose distance is at most ϵ. Let

$$\beta_0(\epsilon, t) = \#\pi_0(\mathcal{R}_\epsilon(X_t))$$

be the number of connected components at resolution ϵ. A clustering velocity profile is a function such as

$$v_t(\epsilon) = -\frac{d}{d \log \epsilon} \log \beta_0(\epsilon, t),$$

or an empirical finite-difference version thereof.

Meaning space is not a flat cloud; it has nested structure, with clusters merging gradually as one passes to coarser levels of semantic resolution. This is the semantic analogue of the closure-system account of hierarchical grammatical universals (Theorem **??**). For grammar, hierarchy universals say that marked features imply less-marked features. For semantic evolution, the hierarchy appears as a multiresolution filtration; the invariant is the stable shape of the cluster-merging profile rather than a single implication.

Proposition 10.4 (Semantic hierarchy as multiresolution closure)**.** *Suppose a lexical-semantic trajectory $L_\bullet$ is generated by a frame hierarchy H such that each frame $h \in H$ contributes a locally coherent semantic region and the frame hierarchy is approximately nested under abstraction. Then the induced filtration $\{\mathcal{R}_\epsilon(X_t)\}_\epsilon$ has a cluster-merging profile whose large-scale component structure is stable under perturbations that preserve the frame hierarchy.*

Proof sketch. Nested frames generate regions whose within-region distances are smaller, in expectation, than between-region distances at the corresponding abstraction level. Perturbations preserving this separation may change exact neighbor relations but do not change the order in which the major regions merge except within the allowed distortion. Hence the resulting β_0-profile is stable up to the induced Q-defect. □

The proposition is not a theorem about all embeddings. It states the structural condition under which the empirical clustering-velocity regularity is exactly what the cognition-language correspondence predicts.

10.6 Persistent temporal dynamics as context-indexed semantic growth

The third regularity is persistent local temporal dynamics: new entities tend to appear disproportionately near other recently created entities. Let $R \subseteq X_t$ be a semantic region and let $N_R(t, \Delta)$ be the number of entities born in R during the interval $[t, t + \Delta]$. A minimal self-exciting model has intensity

$$\lambda_R(t) = \lambda_R^0 + \alpha \sum_{s<t} K(R, R_s) \exp(-\gamma(t - s)),$$

where R_s is the region in which a prior birth occurred, K measures semantic proximity between regions, $\alpha \geq 0$ is a persistence coefficient, and $\gamma > 0$ is a decay rate. The empirical claim is that α is positive for meaningful semantic partitions.

This is the semantic analogue of context-indexed local laws (Theorem **??**). In the grammar case, a genuine law often has the form

$$X \Rightarrow Y \quad \text{inside context } c,$$

rather than globally. In the semantic-growth case, a genuine law has the form

$$\text{recent activity in region } R \quad \Rightarrow \quad \text{increased future birth intensity near } R.$$

The cognitive pullback is straightforward: when a cognitive-cultural frame is active, it keeps generating nearby distinctions for a while, and observed lexical births are the external trace of this frame-local update process. In causal-coding language, context c activates a support S_c; repeated local updates to S_c externalize as a burst of lexical-semantic growth in the corresponding semantic region.

10.7 Taylor's law as a semantic growth-scaling universal

The fourth regularity is Taylor's law: across semantic regions and time windows, the variance of new-entity counts scales as a power of the mean. For a family of regions $\mathcal{R}$, write

$$m_R = \mathbb{E}[N_R], \qquad s_R^2 = \mathrm{Var}(N_R).$$

Taylor's law has the form

$$s_R^2 \approx a m_R^\beta,$$

for constants $a > 0$ and $\beta > 0$. When β differs from the value expected under simple independent Poisson growth, semantic innovation is not uniform random diffusion; it is bursty, regionally structured, and scale-regular.

This adds a sixth class to the taxonomy of universals developed earlier. We now have:

1. closure universals,
2. factorization and mediator universals,
3. low-frustration coherence universals,
4. graded stability universals,
5. context-indexed local universals,
6. semantic growth-scaling universals.

Growth-scaling universals are not primarily implications among features. They are laws of fluctuation in the production of new distinctions. This matters because cognitive universals need not always appear as exceptionless symbolic laws: some appear instead as scaling laws governing the creation of patterns.

10.8 Directed preferential placement as a weakest causal explanation

The directed preferential placement (DPP) model of [**?**] is especially natural from the standpoint of quantale weakness. It explains several regularities with a small number of generative commitments: new entities are placed in semantic space under the combined influence of popularity, history, and local spatial context.

Let $\mathsf{Model}(U)$ be the category of generative explanations of a family U of semantic-growth regularities. For a model M, define a product-quantale score

$$W(M) = \big(\mathrm{Fit}(M),\ -\mathrm{Cost}(M),\ -\mathrm{DC}(M),\ \mathrm{Gen}(M)\big),$$

where Fit measures agreement with the empirical invariants, Cost measures description or computational cost, DC measures double-counting of dependencies, and Gen measures cross-language or cross-dataset generality. The product order is chosen so that larger fit, smaller cost, smaller double-counting, and larger generality are preferred. A weakest causal explanation of U is then

$$M^* \in \underset{M \in \mathsf{Model}(U)}{\arg\max}\ W(M).$$

In this sense, DPP is attractive not merely because it fits data, but because it is weak. It does not posit a separate mechanism for each language, each cluster, or each historical period. It leaves many microhistories possible while explaining the aggregate invariants. A more baroque model that separately stipulates the four regularities would have worse weakness because it would introduce redundant commitments and additional double-counting.

Proposition 10.5 (Weak causal growth explanations beat correlation piles)**.** *Let M_1 and M_2 be two explanations of the same semantic-growth universal family U. Suppose*

$$\mathrm{Fit}(M_1) \geq \mathrm{Fit}(M_2), \qquad \mathrm{Cost}(M_1) \leq \mathrm{Cost}(M_2),$$

$$\mathrm{DC}(M_1) \leq \mathrm{DC}(M_2), \qquad \mathrm{Gen}(M_1) \geq \mathrm{Gen}(M_2),$$

with at least one inequality strict. Then M_1 strictly dominates M_2 in the product-quantale weakness order.

Proof. Immediate from the product order defining W. The proposition makes explicit the Occam-style selection principle: among models that explain the same regularities, prefer the one introducing fewer independent commitments, fewer redundant dependency channels, and greater cross-context generality. $\square$

The semantic-growth results thus give a concrete empirical instance of the general principle used throughout this paper: weakest causal generative models are preferred over piles of descriptive correlations.

10.9 Relation to causal coding and cognitive architecture

The four semantic-growth regularities also align closely with the causal-coding and HBCML picture.

Frequency assortativity is the lexical trace of attention-weighted cognitive substrate: high-salience frames generate high-frequency semantic neighborhoods, just as heavily reused cognitive modules accumulate many contextual uses.

Clustering velocity is the lexical-semantic trace of hierarchical frame organization, the semantic analogue of grammatical closure structure. Where grammatical hierarchies generate implication closures, semantic hierarchies generate stable multiresolution cluster profiles.

Persistent temporal dynamics is the lexical trace of context-local update. When a cognitive frame is activated by a cultural, practical, or communicative context, nearby distinctions are generated in bursts; the resulting word births remain local in semantic space for a historically extended period.

Taylor's law is the lexical trace of nonuniform resource allocation. A modular attention system does not distribute innovation evenly across all regions; it allocates effort to active supports, producing bursty scaling regularities rather than uniform Poisson growth.

In summary:

$$\begin{aligned}
&\text{frequency assortativity} && \leftrightarrow && \text{attention-weighted reusable substrate,}\\
&\text{clustering velocity} && \leftrightarrow && \text{hierarchical frame organization,}\\
&\text{persistent local dynamics} && \leftrightarrow && \text{context-gated cognitive update,}\\
&\text{Taylor scaling} && \leftrightarrow && \text{bursty allocation of cognitive-cultural effort.}
\end{aligned}$$

This makes semantic evolution a natural testbed for the general cognition-language correspondence. Languages grow new meanings the way a modular, attention-weighted, hierarchical cognitive-cultural system would be expected to grow new distinctions.

10.10 Scope and caveats

The empirical results should not be overinterpreted. Static embeddings collapse polysemy into a single vector per word, and the birth of new senses of existing words is not the same as the birth of new lexical items. Historical timing data are much easier to obtain for some languages than for others. These limitations mean that semantic-growth universals should be treated as evidence about the projection of cognitive-cultural dynamics into available lexical data, not as direct observations of cognitive dynamics themselves.

Nevertheless, the fit with the present framework is strong. The grammar-side claim is that morphosyntactic universals are stable subobjects of the linguistic externalization core. The semantic-side claim is that semantic-evolution universals are stable invariants of the frame-to-lexicon growth process. Their cognitive pullbacks are not merely rules of language but structural regularities of concept formation: attention-weighted salience, hierarchical frame structure, context-local persistence, and bursty scaling of innovation. The final picture becomes:

$$\boxed{\text{morphosyntactic universals are stable constraints on linguistic externalization;}}$$

$$\boxed{\text{semantic-evolution universals are stable constraints on how cognitive-cultural meaning space grows;}}$$

$$\boxed{\begin{array}{c}\text{both are observable projections of cognitive invariants}\\ \text{through the same enriched cognition-language correspondence.}\end{array}}$$

11 Statistical and computational research programme

11.1 Recover the linguistic subobjects

The first empirical task is to represent each candidate universal as a subobject $U \hookrightarrow L$ in the typological TUG core. This requires estimating:

- the feature lattice or poset in which the universal lives;
- whether the universal is closure-like, energy-like, mediated, stability-like, or context-indexed;
- its empirical stability coefficient $\sigma(U)$, corrected for genealogy and geography;
- its local context restrictions.

11.2 Infer cognitive pullbacks

For each stable linguistic subobject U, infer candidate cognitive pullbacks $\Lambda^*(U)$. Since Λ is not directly observed, this is an inverse problem. The weakness criterion selects the cognitive explanation $E \in \mathrm{Exp}(U)$ maximizing $W(E)$.

Empirically, this means comparing models with different cognitive mediators, different frame structures, and different modular decompositions, and penalizing double-counted dependencies. A broad universal should not be accepted as cognitively real merely because multiple correlated features point in the same direction; it should be accepted when a weak mediator explains the dependency better than redundant correlations.

11.3 Estimate the confusion graph

The approximate CCL theorem bounds forgetting and path-dependence by a graph of confusing context pairs. The linguistic analogue is a graph of contexts or feature domains whose interactions produce violations of factorization. The research task is to estimate this graph from typological data:

$$G_{\mathrm{conf}} = (\mathsf{Ctx}, E_{\mathrm{conf}}, \delta).$$

Edges should correspond to mediators or context overlaps. Sparse confusion is predicted by causal coding; dense confusion would undermine the modularity hypothesis.

11.4 Test closure transport

For hierarchical universals, estimate the cognitive-frame closure relation R_C and the linguistic closure relation R_L. The categorical prediction is that

$$\mathrm{cl}_L \approx \Sigma_\Lambda \, \mathrm{cl}_C \, \Lambda^*.$$

This can be tested by checking whether linguistic closure defects correspond to plausible cognitive-frame defects and whether the closure direction matches non-linguistic cognitive accessibility data.

11.5 Test sparse-energy pushforward

For word-order universals, infer a signed interaction graph over linguistic choices and ask whether it can be represented as a sparse pushforward of a controller-side routing graph. The key predictions are:

- the linguistic graph should be sparse;
- strong hubs should correspond to high-traffic externalization choices;
- diachronic transitions should favor lower energy;
- higher-order residuals should be small but not zero.

11.6 Test stability-rate transport

Estimate $\sigma_L(U)$ for linguistic universals and compare with candidate cognitive stability measures. In neural or cognitive models, analogues include:

- resistance to fine-tuning perturbation;
- low commutator defects across contexts;
- low demotion rates from kernel or inner-shell compartments;
- high reuse utility across tasks.

The prediction is that linguistic high-stability universals correspond to high cognitive stability under these measures.

11.7 Use multilingual neural models as an intermediate probe

Large multilingual neural models provide an intermediate system between human typology and implemented cognitive architectures. They can be probed for:

- modular supports for typological features;
- context-conditioned activation routing;
- commutator defects under sequential fine-tuning;
- stability gradients across parameters or circuits;
- sparse mediators for broad cross-module regularities.

The question is not whether current transformers are human brains. The question is whether systems trained on multilingual data develop internal structures that mirror the categorical transport patterns predicted here.

12 Implications for Universal Grammar and AGI

12.1 Principles and parameters revisited

The old principles-versus-parameters distinction becomes sharper in this framework.

Principles are high-stability closure laws: direct images of cognitive hard-kernel invariants.

Parameters are externalization choices: linguistic images of controller-side routing and sparse coherence pressures.

Mediated broad universals are residual bridge structures: weakest contexts linking modules that would otherwise factor.

Failed universals are often outer-shell residue or context-indexed laws whose context index has been forgotten.

This preserves the useful insight of the generative tradition that some constraints are deep, while incorporating the empirical fact that most typological regularities are graded, context-sensitive, and historically filtered.

12.2 A constraint on AGI architectures

If the correspondences are correct, then a human-level language-capable AGI should not merely learn language statistics. It should instantiate, at some level, the categorical structure that makes linguistic universals stable:

- a causally factorized cognitive substrate;
- context-local updates with small commutators;
- semantic frames mediating cognition and language;
- sparse externalization routing;
- graded kernel-shell stability;
- residual mediators for genuine cross-module laws;
- context-indexed local closure and gluing.

An architecture lacking these structures may still imitate language for many purposes, but it should struggle with lifelong language learning, robust transfer, and systematic generalization across contexts unless it recreates the same structure indirectly.

12.3 Why the typological record matters for cognitive architecture

The typological record has been shaped by many languages, cultures, diachronic changes, and communicative contexts. Once corrected for genealogy and area, stable patterns in that record are not just facts about grammar; they are aggregated evidence about the cognitive machinery that can generate and maintain grammar under continual learning. This makes linguistic universals a potentially powerful indirect probe of cognitive architecture.

13 Limitations and open questions

13.1 Which correspondences are already theorems?

The factorization correspondence is the strongest result: the TUG trace-factorization theorem and the categorical CCL commutator theorem are instances of one monoidal fact. The other correspondences are conditional theorems under explicit assumptions. The main technical work ahead is to determine whether those assumptions can be derived from a common, deeper model of cognition-language externalization.

13.2 How unique is the functor Λ?

The functor Λ is not directly observable. In many cases the more general profunctor $\mathcal{P}$ is more realistic. Different cognitive substrates may support the same linguistic universal, and the same cognitive substrate may externalize differently across languages. Weakness helps choose among explanations, but it does not make the inverse problem trivial.

13.3 Functional and historical explanations

The framework does not deny functional pressures, processing constraints, diachronic drift, or historical contact. Instead, it asks how such pressures interact with the categorical structure of cognition-language externalization. A functional pressure may be the reason a particular energy term exists; a historical process may explain why a language occupies a particular local minimum. The categorical claim is about the shape of the space and the transport of invariants through that space.

13.4 Substrate neutrality

The categories defined here are substrate-neutral. They can describe biological cortical columns, artificial modular neural systems, symbolic or neural-symbolic systems, or other cognitive architectures. The human-brain interpretation is a hypothesis about one realization of the structure, not a requirement of the formalism.

13.5 Open mathematical questions

1. Can the frame category Frame_Q be constructed explicitly from a typed metagraph or distinction-metagraph representation, rather than postulated abstractly?
2. Can the externalization functor $\mathcal{E}$ be learned from multilingual corpora and typological databases?
3. Can the adjunction $\mathcal{E} \dashv \mathcal{S}$ be validated empirically by production/comprehension asymmetries?
4. Can the closure transport theorem be strengthened without the reflectivity assumptions used in Theorem **??**?
5. Can the sparse-energy pushforward proposition be derived from a concrete HBCML controller rather than assumed as a bounded-cumulant approximation?
6. Can the residual mediator theorem be turned into a practical statistical test for broad universals?
7. Can quantale weakness be estimated robustly from noisy typological and neural data?

14 Conclusion

The relation between linguistic universals and causal-coding cognition developed in this paper is a quantale-enriched correspondence between cognitive causal modules, semantic frames, and linguistic TUG cores. Stated as a single sentence in two directions:

> linguistic universals are direct images of cognitive invariants
> along a weakest, quantale-enriched externalization correspondence

and conversely

> cognitive universals are weakest pullback explanations
> of stable linguistic subobjects.

The first statement says that stable cognitive structure becomes typological structure when transported through semantic frames and externalized into language. The second says that a stable linguistic universal is evidence for upstream cognitive structure, but only for the weakest such structure needed to explain the data. The weakness clause matters: it is what prevents the theory from multiplying hidden modules, hidden parameters, and redundant correlations beyond necessity.

The strongest theorem is the monoidal transport theorem. Disjoint cognitive supports yield commuting cognitive updates; the cognition-language functor maps this into swap equivalence in the TUG trace quotient; factored cognitive substrates yield factored linguistic cores; and nontrivial broad universals require mediators. This unifies the TUG factorization theorem with the causal-continual-learning commutator theorem.

The other four correspondences take theorem-shaped forms as well. Hierarchical universals arise by closure transport from cognitive-frame closure. Narrow word-order universals arise by sparse-energy pushforward from controller coherence. Graded universalhood arises by stability-rate transport from kernel-shell cognitive dynamics. Context-indexed local laws arise by gluing local closure systems over a context category, with defects bounded by confusion costs.

Quantale weakness supplies the selection principle binding all of this together. TyLAA weakness, causal-coding weakness, and dependence-frontier weakness are not separate metaphors. They are composable layers of one generalized Occam principle: preserve as much admissible indistinction as possible while fitting the data and avoiding unnecessary cost, confusion, and double-counting. This is why causal explanations beat redundant correlation explanations in the framework: they identify the weakest mediator that makes a dependency real.

The framework also extends naturally beyond morphosyntax. Inserting a dynamic lexical-semantic layer between frames and observed linguistic form, the four embedding-space regularities of Guo et al. [**?**] frequency assortativity, clustering velocity, persistent local temporal dynamics, and Taylor's law for new-entity counts are recovered as observable shadows of attention-weighted, hierarchical, context-local, and bursty mechanisms of concept formation. This adds a sixth class of universal (semantic growth-scaling) and gives a second, qualitatively different empirical anchor: the same enriched correspondence governs how cognitive-cultural meaning space *grows* as well as how linguistic form is constrained. Directed preferential placement is preferred over more baroque generative alternatives for exactly the same reason a single mediator is preferred over a pile of correlations: it dominates them in the product-quantale weakness order.

The resulting research programme is clear. Represent universals as linguistic subobjects (or as invariants of growth trajectories). Infer their weakest cognitive pullbacks. Test whether closure, factorization, sparse energy, stability, context-gluing, and growth-scaling structures align across typology, semantic-embedding evolution, cognitive experiments, neural models, and AGI architectures. If they do, the typological and lexical-semantic records become more than catalogs of language facts. They become an indirect map of the cognitive architecture that makes human language possible.

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